

Temporal fingerprints

A logarithmic reference frame of time scales
for gravitational and quantum systems

Frédéric Vronsky

Independent researcher in theoretical cosmology, Toulouse, France
ORCID: 0009-0003-5719-9604

Toulouse — October 2026 · Version 1.1

Licence Creative Commons BY-NC-SA 4.0 International

Abstract

Any physical system characterised by a mass M , a length R and an observed clock T_{obs} defines four time scales: the Planck time $T_P = \sqrt{\hbar G/c^5}$, the gravitational time $T_G = GM/c^3$, the light-crossing time $T_C = R/c$ and the observed time itself; a quantum particle adds its reduced Compton time $T_Q = \hbar/(mc^2)$. We call *temporal fingerprint* this set together with its dimensionless relative state $(X, Y, O) = (GM/Rc^2, M_P/M, cT_{\text{obs}}/R)$. We show that, in the space of the logarithms of these times, (i) a change of unit is a translation that leaves the relative state invariant; (ii) the number of independent coordinates is exactly three (Buckingham's theorem seen as the rank-nullity theorem); (iii) every power law becomes a hyperplane, and physical thresholds (Schwarzschild radius, innermost stable orbit, Planck mass and length, Compton length) cut the space into cells on which a conventional classification of regimes can be based; (iv) eliminating a variable reduces to computing a kernel; (v) the deviation of a system from a law becomes a signed distance, and logarithmic covariances propagate exactly for power laws. Three demonstrators illustrate these properties: Keplerian orbits and compact objects in an atlas of 33 systems; the quantum extension, where the product of the self-gravity and internal-clock phases no longer depends on the branch separation, yielding a purely temporal access condition; and the Hubble sphere, of compactness $X = \Omega_{\text{tot}}/2$ at all epochs. **We do not propose any new dynamics:** we build a system of temporal coordinates in which known laws take a linear form, which makes it possible to classify, compare and audit very different systems in a common language. Code, atlas, a script that symbolically checks every equation and an interactive bench accompany the paper.

Keywords: time scales; dimensional analysis; Planck time; Compton time; compactness; general relativity; mathematical methods.

Contents

1	Introduction	2
2	The fundamental time scales	3
3	The temporal fingerprint and its relative state	4
4	Mathematical structure	5
4.1	The logarithmic space	5
4.2	Scale invariance	5
4.3	Counting: Buckingham's theorem as the rank-nullity theorem	5
4.4	Laws as submanifolds	6
4.5	Regimes as cells of a hyperplane arrangement	6

4.6	Eliminating a variable by computing a kernel	7
4.7	The residual as a signed distance	7
4.8	Propagation of uncertainties	8
4.9	Trajectories	9
4.10	Numerical conditioning	9
5	Identities, degeneracies and null tests	9
6	Uses	9
7	Demonstrators	10
7.1	Orbits and compact objects	10
7.2	Quantum extension	10
7.3	Hubble sphere	12
8	Previous work	12
9	Limitations	13
10	Conclusion	13
A	Temporal atlas (excerpt)	14

1 Introduction

Time enters physics in several forms that do not resemble each other. A mass defines a gravitational time, a length a light-propagation time, a particle a quantum time, and the fundamental constants together a Planck time. Observations, for their part, supply real clocks: orbital periods, pulsar rotations, gravitational-wave frequencies, durations of quantum superpositions. These quantities are rarely placed side by side, because they belong to fields (celestial mechanics, relativity, quantum physics, cosmology) that use different variables.

This paper builds a *common reference frame*: each system is described only by its time scales, then by their ratios, then by the logarithms of these ratios. It asks a simple question: what do we gain, mathematically, by expressing everything in time?

What the paper does and does not do. The formalism modifies neither general relativity nor quantum mechanics. The transformations introduced are changes of variables, and every relation they reveal follows from known laws. Its interest must therefore come from what the representation makes visible, classifies, compares or allows one to compute more simply. Throughout the text we carefully distinguish *definitions*, *identities* (true by construction), *derived results* (mathematical consequences of known laws) and *measurements*.

Contributions.

1. Complete definitions of the five time scales, with origin, dimension, domain and limits of interpretation (section 2).
2. The temporal fingerprint and its relative state (X, Y, O) , with their remarkable products (section 3).
3. The mathematical structure of the logarithmic space: scale invariance, counting, laws as hyperplanes, regimes as cells, elimination by kernel, residuals as distances, propagation of logarithmic covariances (section 4).

4. An explicit classification of identities and circularity traps (section 5).
5. Three demonstrators and a temporal atlas of 33 systems, from the proton to the Hubble sphere (section 7 and appendix A).

2 The fundamental time scales

Each scale is presented with the same scheme: definition, origin, dimension, interpretation, domain, and what it is not. Constants are those of CODATA 2022 [1], whose values of c , $\hbar$ and G are identical to those of 2018; c and $\hbar$ are exact, and the relative uncertainty of G (2.2×10^{-5}) propagates to T_P , M_P and T_G . Whenever possible we use the gravitational parameter $\mu = GM$ directly, which is known far more precisely than G and M separately: this is the case for the Sun, for which we adopt the IAU nominal value $GM_\odot = 1.3271244 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$ [2].

Light-crossing time T_C

Definition. $T_C(R) \equiv \frac{R}{c}$.

Origin. For local light propagation, $\Delta x = c \Delta t$; T_C is the time light needs to cross the scale R , so $c T_C = R$.

Dimension. $[R] = \text{L}$, $[c] = \text{L T}^{-1}$, hence $[T_C] = \text{T}$.

Interpretation. A length expressed as a time. The choice of R (radius, semi-major axis, separation, branch separation) is part of the definition of the system and must be documented.

Domain. Any length. In curved spacetime, R is a coordinate or a proper distance depending on context, which must be specified.

What it is not. It is not a “time of space” nor the proper time of a clock: it is a scale built from a length.

Gravitational time T_G

Definition. $T_G(M) \equiv \frac{GM}{c^3}$.

Origin. The gravitational radius of a mass is $r_g = GM/c^2$; its light-crossing time is $r_g/c = GM/c^3$. Thus $c T_G = r_g$, and the Schwarzschild radius $r_s = 2r_g$ satisfies $r_s/c = 2T_G$.

Dimension. $[G] = \text{L}^3 \text{M}^{-1} \text{T}^{-2}$ hence $[GM/c^3] = \text{T}$.

Interpretation. A mass expressed as a time. It is the usual mass unit of pulsar timing, where $T_G(M_\odot) = 4.925491 \text{ } \mu\text{s}$.

Domain. Any mass. Its precise relativistic meaning depends on the geometry considered.

What it is not. It is not a time “created” by the mass, nor in general the proper time of a clock.

Planck time T_P

Definition. $T_P \equiv \sqrt{\frac{\hbar G}{c^5}} \simeq 5.391246 \times 10^{-44} \text{ s}$.

Origin. We look for $T_P = \hbar^\alpha G^\beta c^\gamma$ with dimension T . With $[\hbar] = \text{ML}^2 \text{T}^{-1}$, equating the exponents of M , L and T gives the system $\alpha - \beta = 0$, $2\alpha + 3\beta + \gamma = 0$, $-\alpha - 2\beta - \gamma = 1$, whose unique solution is $\alpha = \beta = \frac{1}{2}$, $\gamma = -\frac{5}{2}$. One deduces the length $L_P = c T_P$ and the mass $M_P = \sqrt{\hbar c / G}$.

Property. $T_G(M_P) = GM_P/c^3 = T_P$: the Planck time is the gravitational time of the Planck mass.

Interpretation. The unique monomial $\hbar^\alpha G^\beta c^\gamma$ with the dimension of a time: any other time scale built from these constants alone equals it up to a dimensionless factor. It depends on no system and serves as a common anchor.

What it is not. It is neither a proven minimal duration nor a prediction about the structure of time: it is a unit.

Reduced Compton time T_Q

Definition. $T_Q(m) \equiv \frac{\hbar}{mc^2}$, for a particle of mass m .

Origin. The phase of a particle at rest evolves as $e^{-imc^2\tau/\hbar}$: T_Q is the inverse of its proper angular frequency. It is also $\bar{\lambda}_C/c$, with $\bar{\lambda}_C = \hbar/(mc)$ the reduced Compton length. The time associated with the ordinary Compton frequency, defined with $h = 2\pi\hbar$, is 2π times larger; all relations in this paper use the reduced form.

Property. $T_G(m)T_Q(m) = \frac{Gm}{c^3} \frac{\hbar}{mc^2} = T_P^2$: the Planck time is the geometric mean of the gravitational and quantum times of the same object.

What it is not. It is not a directly measurable duration; it fixes the conversion between proper time and quantum phase.

Observed time T_{obs}

Definition. $T_{\text{obs}} \equiv$ time scale attached to the observable under study.

Realisations. Orbital period P_{orb} (Keplerian: $2\pi\sqrt{R^3/GM}$); inverse gravitational-wave frequency $1/f_{\text{GW}}$; pulsar spin period; duration of a quantum superposition; cosmic age.

Difference in nature. T_P, T_G, T_C and T_Q are *constructed* from constants and parameters; T_{obs} is *attached to a measurement*. All the testing capacity of the formalism goes through T_{obs} .

Requirement. The nature of T_{obs} and its independence from M and R must always be documented (section 5).

3 The temporal fingerprint and its relative state

Definition 1 (Temporal fingerprint). The temporal fingerprint of a system is the set

$$\mathcal{E} = [T_P, T_G, T_C, T_{\text{obs}} ; \mathcal{T}_{GCP}^{\text{rel}}], \quad (1)$$

made of an *absolute level* (the times, in seconds) and a dimensionless *relative state*

$$\mathcal{T}_{GCP}^{\text{rel}} = (X, Y, O; F_{\mathcal{M}}), \quad X \equiv \frac{T_G}{T_C} = \frac{GM}{Rc^2}, \quad Y \equiv \frac{T_P}{T_G} = \frac{M_P}{M}, \quad O \equiv \frac{T_{\text{obs}}}{T_C} = \frac{cT_{\text{obs}}}{R}, \quad (2)$$

where $F_{\mathcal{M}}$ denotes the proper-time factor given by a relativistic model $\mathcal{M}$ explicitly chosen for the system (Schwarzschild, Kerr, post-Newtonian expansion, pulsar timing model, etc.). The coordinates (X, Y, O) are common to all systems; $F_{\mathcal{M}}$ depends on the model.

Equation (2) summarises the tool in one line: X is the compactness, Y the (inverse) mass in Planck units, O the observed clock in light-crossing units.

Remarkable products. Simple combinations of X and Y recover the usual scales:

$$XY = \frac{T_P}{T_C} = \frac{L_P}{R}, \quad (3)$$

$$XY^2 = \frac{T_Q}{T_C} = \frac{\bar{\lambda}_C}{R}, \quad (4)$$

$$X^2Y = \frac{GM/R^2}{a_P}, \quad a_P \equiv \frac{c}{T_P} \simeq 5.56 \times 10^{51} \text{ m s}^{-2}. \quad (5)$$

Gravitation (X), mass (Y), geometry (XY), quantum (XY^2) and acceleration (X^2Y) are expressed in the same system of ratios. These are identities, but their common organisation is what constitutes the tool.

Relativistic factors. In the model $\mathcal{M} = \text{Schwarzschild}$, for a Schwarzschild metric, a static clock at R and a clock on a circular geodesic orbit tick, relative to time at infinity, as

$$F_{\text{stat}} = \sqrt{1 - 2X}, \quad F_{\text{orb}} = \sqrt{1 - 3X}. \quad (6)$$

Remark. These expressions hold outside a spherical, uncharged, non-rotating body. They are not a universal formula for the relative state. The innermost stable circular orbit lies at $X = 1/6$, the horizon at $X = 1/2$.

4 Mathematical structure

4.1 The logarithmic space

Let $t = (\ln T_P, \ln T_G, \ln T_C, \ln T_{\text{obs}})^\top \in \mathbb{R}^4$. Ratios become differences and the relative state is the image of t under a linear map:

$$\begin{pmatrix} \ln X \\ \ln Y \\ \ln O \end{pmatrix} = A t, \quad A = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \quad (7)$$

4.2 Scale invariance

Proposition 1. *A change of time unit $T_i \mapsto \lambda T_i$ acts on t by the translation $t \mapsto t + (\ln \lambda) \mathbf{1}$ and leaves (X, Y, O) invariant.*

Proof. Each row of A sums to zero, so $A\mathbf{1} = 0$ and $A(t + \mu\mathbf{1}) = At$. $\square$

The relative state is therefore a coordinate system on the quotient space $\mathbb{R}^4/\mathbb{R}\mathbf{1}$, of dimension 3. This is a proven property of the reference frame, not a physical assumption.

4.3 Counting: Buckingham's theorem as the rank-nullity theorem

The variables of a system are $(M, R, T_{\text{obs}}, G, c, \hbar)$. Their dimension matrix, with rows (M, L, T), is

$$D = \begin{pmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 1 & 2 \\ 0 & 0 & 1 & -2 & -1 & -1 \end{pmatrix}. \quad (8)$$

Proposition 2. *D has rank 3. The kernel of D , which parametrises the dimensionless groups $M^a R^b T_{\text{obs}}^c G^d c^e \hbar^f$, has dimension $6 - 3 = 3$, and the exponent vectors of X , Y and O ,*

$$X : (1, -1, 0, 1, -2, 0), \quad Y : (-1, 0, 0, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2}), \quad O : (0, -1, 1, 0, 1, 0),$$

form a basis of it.

Proof. The columns of M , R and T_{obs} form the identity matrix, so the rank is 3. One checks directly that $Dx = Dy = Do = 0$, and the three vectors are independent (their components on M , R , T_{obs} form an invertible matrix). The rank-nullity theorem gives the dimension of the kernel. $\square$

Any other combination ($Z = T_P/T_C$, T_{obs}/T_G , T_{obs}/T_P , etc.) is a product of powers of X , Y and O ; for instance $Z = XY$. Adding T_Q of the same mass brings nothing, since $\ln T_Q = 2\ln T_P - \ln T_G$. Adding the Compton time of a second mass m adds one dimension.

Proposition 3. *Any dimensionless relation between M , R , T_{obs} , G and c , i.e. one not involving $\hbar$, can be written with X and O alone. Any relation between these same variables and $\hbar$ that involves $\hbar$ necessarily involves Y ; if G is absent from it, Y appears only through the combination $XY^2 = \bar{\lambda}_C/R$.*

Proof. Without $\hbar$, the dimension matrix of the five variables $(M, R, T_{\text{obs}}, G, c)$ has rank 3; its kernel has dimension 2 and is spanned by the exponent vectors of X and O , whose $\hbar$ component is zero. The vector of Y has a component $\frac{1}{2}$ on $\hbar$: any group in which $\hbar$ has a non-zero exponent therefore contains Y . Without G , the kernel associated with $(M, R, T_{\text{obs}}, c, \hbar)$ is spanned by O and by $\hbar/(McR) = XY^2$, whose G component is zero. $\square$

Since vacuum general relativity does not contain $\hbar$, its relations depend only on X and O : this is how the absence of an intrinsic mass scale in classical gravitation appears in this frame. An additional variable, such as the energy splitting ΔE of an internal clock (section 7.2), introduces a new scale $\hbar/\Delta E$; the proposition then applies to the enlarged set of variables.

4.4 Laws as submanifolds

Every power law $\prod q_i^{n_i} = K$ between quantities of the frame becomes, after taking logarithms, the affine hyperplane $\sum n_i \ln q_i = \ln K$.

Example 1 (Kepler). The third law $T_{\text{obs}} = 2\pi\sqrt{R^3/GM}$ reads $O\sqrt{X} = 2\pi$, i.e. the hyperplane

$$\ln O + \frac{1}{2} \ln X - \ln(2\pi) = 0. \quad (9)$$

For a quasi-circular gravitational wave, $f_{\text{GW}} = 2f_{\text{orb}}$ and the constant becomes π .

Physical thresholds are hyperplanes too: the horizon $X = 1/2$, the innermost stable orbit $X = 1/6$, the Planck mass $Y = 1$, the Planck length $XY = 1$ and the Compton length $XY^2 = 1$. Laws that are not power laws give curved manifolds: $\ln F_{\text{stat}} = \frac{1}{2} \ln(1 - 2X)$ is not linear in $\ln X$. In the weak field, $\ln F_{\text{stat}} \simeq -X$: the manifold is flat to order X , which naturally delimits the Newtonian domain.

Figure 1 is the central illustration of this property: in the $(\ln X, \ln O)$ plane, a law becomes a line and each system with a clock a point whose distance to that line can be measured. It is the O axis, absent from mass-size diagrams, that makes this reading possible.

4.5 Regimes as cells of a hyperplane arrangement

The preceding hyperplanes divide the space $(\ln X, \ln Y, \ln O)$ into convex cells. Each cell corresponds to a regime, according to a classification that we choose and that has no status of universal boundary: inside the Schwarzschild radius ($X > 1/2$), the zone between the innermost stable orbit and that radius ($1/6 < X < 1/2$, in the Schwarzschild interpretation), weak field ($X \ll 1$), sub-Planckian mass ($Y > 1$), length below the Compton length ($XY^2 > 1$), etc. Figure 2 shows the projection onto the $(\ln X, \ln Y)$ plane: the Schwarzschild, Compton and Planck lines meet there at the Planck point $(X, Y) = (1, 1)$. Classifying a system amounts to finding the cell that contains it, which is a sign computation.

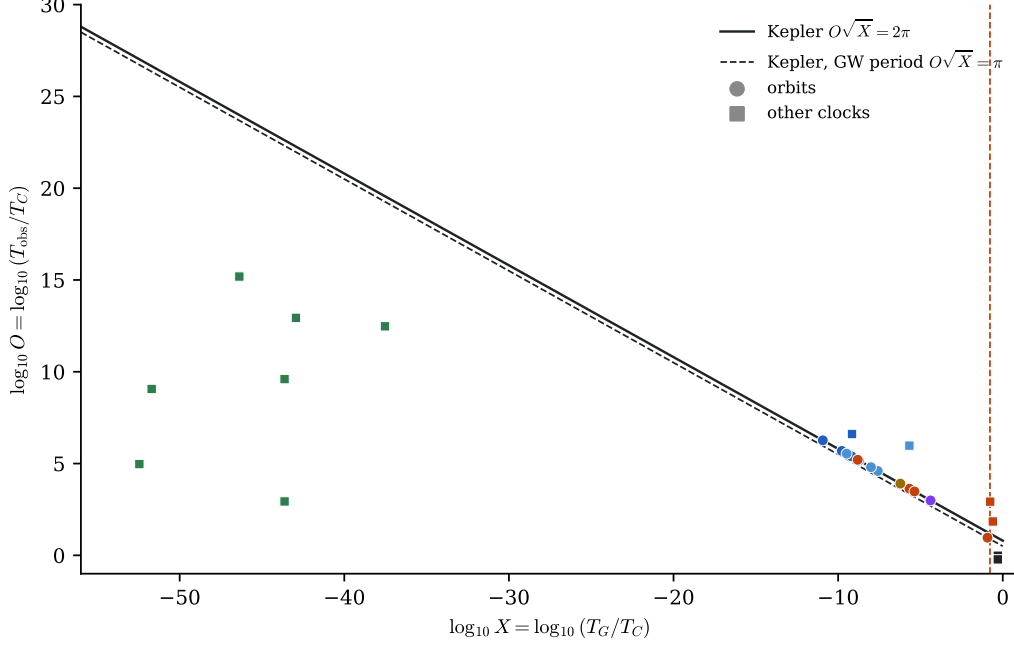


Figure 1: Central figure of the frame: the clock plane $(\log_{10} X, \log_{10} O)$. Orbits lie on the Kepler hyperplane $O\sqrt{X} = 2\pi$; the GW150914 coalescence, whose clock is the wave period, lies on $O\sqrt{X} = \pi$. The dotted vertical line marks the innermost stable orbit.

4.6 Eliminating a variable by computing a kernel

Consider log-linear quantities $\ln q_i = \sum_j C_{ij} \ln p_j$ in parameters p_j .

Proposition 4. *A product $\prod_i q_i^{w_i}$ is independent of the parameters p_j , $j \in J$, if and only if w belongs to the kernel of $C_J^\top$, where C_J is the restriction of C to the columns J .*

Proof. $\ln \prod q_i^{w_i} = \sum_j (\sum_i w_i C_{ij}) \ln p_j$; the coefficient of $\ln p_j$ vanishes for all $j \in J$ exactly when $C_J^\top w = 0$. $\square$

Building invariants thus becomes systematic; section 7.2 gives an example in which the separation between the branches of a superposition disappears.

4.7 The residual as a signed distance

For a hyperplane $\Pi : n \cdot p = b$ of the logarithmic space, the residual of a system p is

$$r_\Pi(p) = \frac{n \cdot p - b}{\|n\|}. \quad (10)$$

For Kepler we use the non-normalised form $r_K = \ln O + \frac{1}{2} \ln X - \ln 2\pi$, which reads directly as $r_K = \ln(T_{\text{obs}}/T_{\text{Kepler}})$, the relative difference between the observed and the predicted period. We write $K_T \equiv O\sqrt{X}/2\pi = T_{\text{obs}}/T_{\text{Kepler}}$ for the ratio to the template, $A_T \equiv K_T - 1$ for the relative deviation and $r_T \equiv \ln K_T = r_K$ for the logarithmic residual, which treats a factor 2 and a factor 1/2 symmetrically; for a gravitational-wave clock, 2π becomes π . A Euclidean distance in the logarithmic space is a *relative error*, which suits quantities spread over tens of orders of magnitude. For a curved manifold $\mathcal{M}_F : F(p) = 0$, the residual is the distance to $\mathcal{M}_F$, approximated to first order by $F(p)/\|\nabla F(p)\|$.

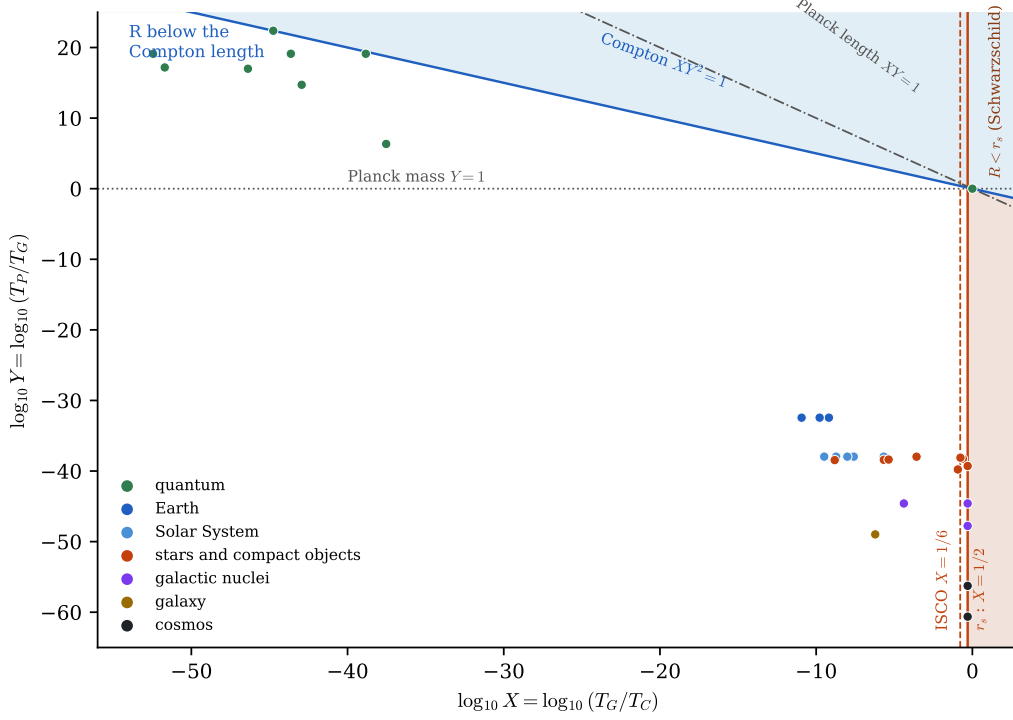


Figure 2: Projection of the atlas onto the $(\log_{10} X, \log_{10} Y)$ plane. The hyperplanes of the horizon, innermost stable orbit, Planck mass, Planck length and Compton length divide the regimes; they meet at the Planck point. This plane is the affine image of the mass-size diagram of Carr and Rees [7] and of Lineweaver and Patel [8] (section 8).

Methodological chain. The preceding sections combine into a single procedure:

$$\text{system} \longrightarrow \text{fingerprint} \longrightarrow (\ln X, \ln Y, \ln O) \longrightarrow \text{theoretical manifold} \longrightarrow \text{residual}.$$

This chain, rather than the mere expression in seconds, is what defines the tool.

4.8 Propagation of uncertainties

Let Σ_t be the covariance matrix of the logarithms of the inputs. For any log-linear map with (constant) Jacobian J ,

$$\Sigma_{\text{out}} = J \Sigma_t J^\top, \quad (11)$$

without approximation: the propagation of logarithmic covariances is exact for log-linear transformations. Converting a usual uncertainty σ_T into $\sigma_{\ln T}$ is exact for log-normal errors and gives $\sigma_{\ln T} \simeq \sigma_T/T$ for small relative errors; for large uncertainties of arbitrary shape it is an approximation. For a general law, J is the matrix of *elasticities* $\partial \ln f / \partial \ln p$, which directly measure by how many percent the result changes when an input changes by 1%.

Example 2. Condition (15) of section 7.2, $T_{\text{obs}} \geq (M_P/m) \sqrt{(\hbar/\Delta E)(c/g)}$, is log-linear: its Jacobian with respect to $(\ln m, \ln \Delta E, \ln g)$ is $J = (-1, -\frac{1}{2}, -\frac{1}{2})$. Independent relative uncertainties of 10% on the mass and 20% on the clock energy splitting therefore give on the minimal duration $\sigma_{\ln T} = \sqrt{0.1^2 + (0.2/2)^2} \simeq 0.14$, with no approximation other than converting uncertainties into logarithmic deviations. Elasticities read directly: a clock twice as energetic reduces the duration by a factor $\sqrt{2}$.

4.9 Trajectories

An evolving system traces a curve. An inspiralling compact binary slides along the Kepler hyperplane (9) (constant π for the wave clock) towards $X = 1/6$, where the Keplerian approximation stops making sense; for GW150914 this happens near $f_{\text{GW}} \simeq 67$ Hz. The Hubble sphere, for its part, stays at $X = 1/2$ for a flat universe (section 7.3).

4.10 Numerical conditioning

The times in the atlas range from 10^{-44} to 10^{18} s. Their logarithms are numbers of order 100, which removes overflow problems and makes comparisons, averages and fits numerically stable.

5 Identities, degeneracies and null tests

Table 1 classifies the relations encountered. It tells the reader in advance what does *not* constitute a discovery.

Relation	Status	Comment
X, Y, O	definitions	changes of variables
$Z = XY = L_P/R, XY^2 = \bar{\lambda}_C/R,$ $X^2Y = g/a_P$	identities	products of coordinates
$T_G(M_P) = T_P, T_G T_Q = T_P^2$	identities	consequences of the definitions
$O\sqrt{X} = 2\pi$	known law	Kepler, hyperplane (9)
$X = \Omega_{\text{tot}}/2$ for the Hubble sphere	derived result	consequence of the Friedmann equation
$\varphi_G \varphi_{\text{clock}}$ independent of Δx	derived result	proposition 4
$r_K \simeq 0$ for the planets	measurement	but $GM_\odot$ is calibrated on these orbits

Table 1: Status of the relations.

Circularity traps. A residual is a test only if the quantities compared are independent.

- If R is reconstructed from T_{obs} through Kepler’s law, then $r_K = 0$ by construction. The atlas shows it: for binary pulsars, S2 or Sirius A–B, $r_K \sim 10^{-15}$, i.e. zero to numerical precision.
- If M is derived from the same orbit (the Sun around the Galactic centre, with $M = V^2 R/G$), the residual is again zero by construction.
- For the planets, a and P are measured independently, but $GM_\odot$ is itself fitted to the planetary ephemerides. The observed residuals ($|r_K| \lesssim 10^{-3}$) mainly reflect data rounding and the neglected planetary mass ($r_K \simeq -q/2$ for a mass ratio q), not a deviation from the law.

When does the tool bring something? We retain four criteria: (1) it allows comparing regimes usually expressed in different variables; (2) it produces a residual with respect to a theoretical manifold, with independent inputs; (3) a regularity persists after eliminating algebraic identities; (4) it leads to a testable condition. The demonstrators below illustrate (1), (2) and (4); none claims to satisfy (3) in the sense of a new law.

6 Uses

Classify. Place systems from different fields on the same map and read their regime (figure 2).

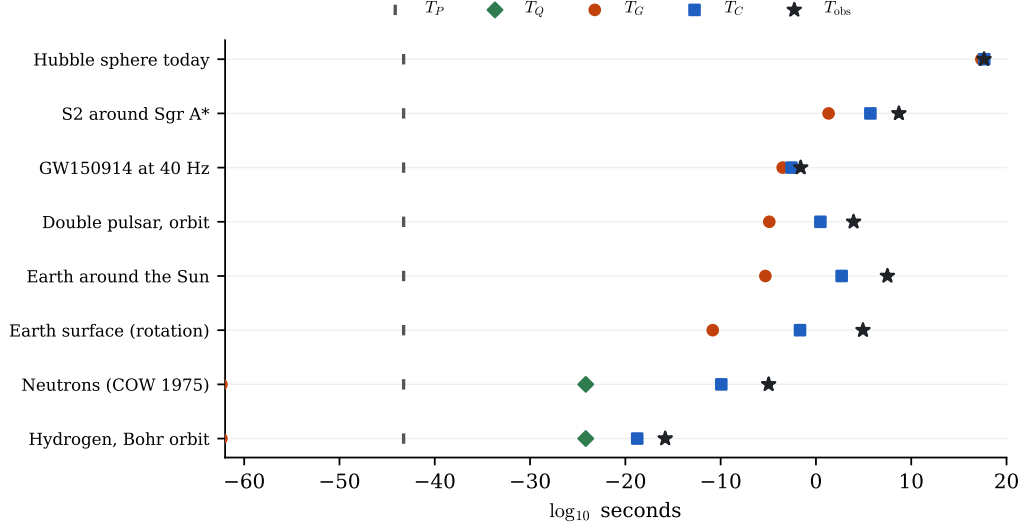


Figure 3: Absolute level of the fingerprint for eight systems of the atlas. The Planck time serves as a common anchor; the gaps between the marks are the relative state.

Audit. Detect a hidden dependence, i.e. a circularity: if a “measured” quantity is in fact a function of the others, the rank of the system of equations reveals it (section 5).

Design. Choose the parameters of an experiment to reach a given cell of the map (section 7.2).

Propagate. Compute the uncertainties and sensitivities of a result (section 4).

Teach. Read at a glance orders of magnitude spread over more than sixty decades (figure 3).

7 Demonstrators

7.1 Orbits and compact objects

Figure 1 (section 4) places the systems that have a clock in the $(\ln X, \ln O)$ plane. Orbits, from the International Space Station to the star S2 around Sgr A*, line up on the Kepler hyperplane over more than five decades of compactness; the GW150914 coalescence at 40 Hz lies on the hyperplane with constant π . Clocks of another nature (Earth or Sun rotation, pulsar spins, quantum superposition durations) sit elsewhere, as expected: only an orbital clock must satisfy (9). The surface of a neutron star ($X \simeq 0.25$ for PSR J0740+6620) and the horizons ($X = 1/2$) appear in the strong-field cell.

This figure shows what the O axis adds with respect to mass–size diagrams: one no longer reads only the position of an object, but how its clock compares with the law. Section 5 states when this residual is a test.

7.2 Quantum extension

Consider a particle of mass m in a superposition of two branches separated by Δx during T_{obs} , in an external field g , carrying an internal clock with energy splitting ΔE . Two dimensionless phases characterise the situation.

Self-gravity of the superposition.

$$\varphi_G \equiv \frac{T_{obs}}{Y^2 T_C(\Delta x)} = \frac{G m^2 T_{obs}}{\hbar \Delta x}. \quad (12)$$

Here Δx is the separation between the two branches of a *single* superposition. The combination $\hbar\Delta x/(Gm^2) = Y^2 T_C(\Delta x)$ is, up to a factor of order unity and for a separation large compared with the size of the object, the time scale of Diósi–Penrose collapse [9, 10], $\hbar/E_G$ with $E_G \simeq Gm^2/\Delta x$. The gravitationally mediated entanglement phase targeted by the proposals of Bose et al. and Marletto and Vedral [11, 12] has the same form, $T_{\text{obs}}/(Y^2 T_C(d))$, but the length there is the distance d between *two* distinct masses, not Δx . These two phenomena are distinct: their similarity of form does not make them a single physical boundary. We therefore speak of the region where the self-gravity of the superposition becomes dynamically relevant.

Distinguishable proper time. The proper-time difference between two branches separated vertically by Δx is at most $\Delta\tau = T_{\text{obs}} g \Delta x/c^2$, and the phase of the internal clock is

$$\varphi_{\text{clock}} \equiv \frac{\Delta E \Delta\tau}{\hbar}. \quad (13)$$

When $\varphi_{\text{clock}} \gtrsim 1$, proper time writes which-path information into the clock and reduces the interference contrast [13]. In a closed free-fall interferometer, the proper-time difference cancels to first order; dedicated geometries are needed to isolate it [14]. The values of $\Delta\tau$ below are therefore upper bounds.

Proposition 5. *The product of the two phases does not depend on the branch separation:*

$$\varphi_G \varphi_{\text{clock}} = \frac{T_{\text{obs}}^2}{Y^2 T_{\text{clock}} T_g}, \quad T_{\text{clock}} \equiv \frac{\hbar}{\Delta E}, \quad T_g \equiv \frac{c}{g}. \quad (14)$$

Consequently, both phases can reach unity simultaneously if and only if

$$T_{\text{obs}} \geq Y \sqrt{T_{\text{clock}} T_g}, \quad (15)$$

the separation Δx then being fixed at $\Delta x_{\text{eq}} = \hbar c^2/(\Delta E g T_{\text{obs}})$.

Proof. In logarithms, $\ln \varphi_G = 2 \ln m - \ln \Delta x + \ln T_{\text{obs}} + \text{const}$ and $\ln \varphi_{\text{clock}} = \ln \Delta x + \ln T_{\text{obs}} + \ln \Delta E + \ln g + \text{const}$. The vector $w = (1, 1)$ cancels the coefficient of $\ln \Delta x$ (proposition 4). The product equals $Gm^2 \Delta E g T_{\text{obs}}^2/(\hbar^2 c^2)$; using $Gm^2/(\hbar c) = Y^{-2}$ yields (14). Varying Δx moves the point along a line of slope -1 in the $(\ln \varphi_G, \ln \varphi_{\text{clock}})$ plane; this line crosses the quadrant $\{\varphi_G \geq 1, \varphi_{\text{clock}} \geq 1\}$ if and only if the product is at least 1. $\square$

In the Earth’s field, $T_g = c/g \simeq 3.06 \times 10^7$ s, about one year. Table 2 gives the minimal duration for a few realistic choices. Figure 4 places the experiments performed and the proposal of Bose et al. (with, for illustration, a 2 eV optical clock): each becomes a segment of slope -1 traced by varying Δx . The experiments performed occupy quadrant Q1; the Bose et al. proposal reaches the boundary of Q3 (self-gravity of its own 250 μm superposition: $\varphi_G \simeq 0.63$; entanglement phase evaluated at the distance d between the two masses: 0.79 at 200 μm , 0.35 at 450 μm) but remains four decades away from Q4 for the product (1.3×10^{-4}).

Mass and internal clock	T_{clock}	minimal T_{obs}	Δx_{eq}
10^{-14} kg, NV spin (2.87 GHz)	5.5×10^{-11} s	9.0×10^4 s (25 h)	5.7 m
10^{-14} kg, optical clock (2 eV)	3.3×10^{-16} s	218 s	1.4 cm
10^{-14} kg, thorium-229 nuclear transition (8.4 eV)	7.8×10^{-17} s	107 s	6.7 mm
10^{-12} kg, optical clock (2 eV)	3.3×10^{-16} s	2.2 s	1.4 m

Table 2: Condition (15) in the Earth’s field. The frequency of the internal clock matters as much as the mass: going from a microwave to an optical clock divides the required duration by about 400.

In this extension, Y stops being a mere unit: it enters only φ_G , i.e. when the object is both quantum and a source of gravity, whereas φ_{clock} depends on neither m nor $\hbar$ except through the clock. This separation illustrates proposition 3: in the demonstrators considered here, X and O suffice to express the classical gravitational relations, while Y becomes explicitly necessary in relations involving $\hbar$.

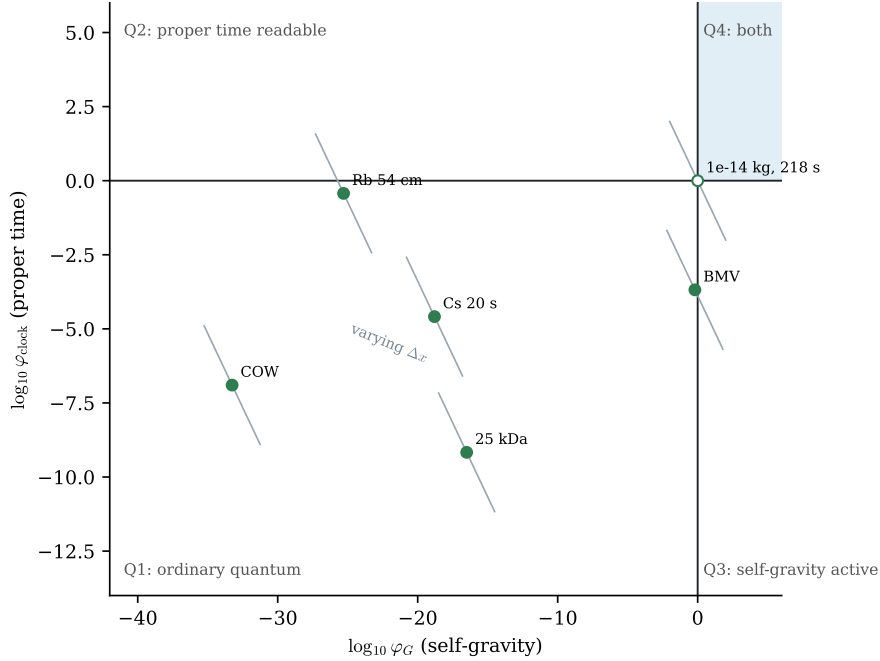


Figure 4: Quadrants of the self-gravity and clock phases (2 eV optical clock assumed, Earth’s field). Each grey segment corresponds to a variation of Δx over four decades. The open point is a hypothetical 10^{-14} kg system held for 218 s, at the entrance of quadrant Q4.

7.3 Hubble sphere

The Hubble sphere. Take $R = c/H$ and the enclosed mass $M = \frac{4\pi}{3}R^3\rho$, with $\rho = \Omega_{\text{tot}} 3H^2/(8\pi G)$ from the Friedmann equation. Then

$$X_{\text{Hubble}} = \frac{GM}{Rc^2} = \frac{\Omega_{\text{tot}}}{2}, \quad (16)$$

at all epochs. For a flat universe, the Hubble sphere therefore has the same numerical value of compactness, $X = 1/2$, as the Schwarzschild threshold, although the two surfaces do not have the same geometric nature: the Hubble sphere of a Friedmann universe is not a Schwarzschild horizon. Lineweaver and Patel make a related remark in the mass–size diagram [8]. With $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and an age of 13.79 Gyr [19], the observed clock is $O = H_0 t_0 \simeq 0.95$.

8 Previous work

The idea of reducing everything to time is not new, and the formalism builds on several traditions.

- **Natural units and dimensional analysis.** Planck introduced in 1899 the units built on $\hbar$, G and c [3]; Buckingham’s theorem [4] underlies the counting of dimensionless groups.
- **Geometrised units.** In general relativity and pulsar timing, masses are commonly expressed in seconds through GM/c^3 ; this is our T_G .
- **Chronometry.** Synge showed that in relativity geometry can be defined from clocks alone [5].
- **Metrology.** Since 2019, the International System of Units defines the metre and the kilogram from the second, through c and h [6].
- **Diagrams of all objects.** Carr and Rees [7], then Lineweaver and Patel [8], place all known objects in a mass–size plane bounded by the Schwarzschild and Compton lines, which cross at the Planck point.

Relation to the mass–size diagram. The $(\ln X, \ln Y)$ plane is the image of the $(\ln M, \ln R)$ plane under the invertible affine map $\ln X = \ln(G/c^2) + \ln M - \ln R$, $\ln Y = \ln M_P - \ln M$. It therefore contains exactly the same information: we do not present it as a new construction. What the present work adds is (i) the observed-clock axis O , which turns a static map into an instrument for comparison with laws; (ii) an entirely temporal formulation, quantum extension included; (iii) the algebraic structure of section 4. We are not aware of a compilation of physical systems organised in this way, but this statement remains to be confronted with the literature.

9 Limitations

- The formalism brings no new physical law; its relations are definitions, identities or consequences of known laws.
- The definition of R and T_{obs} is the modeller’s choice and conditions every interpretation; a poor definition can manufacture a regularity.
- The relativistic factors used hold for the Schwarzschild geometry; rotating, charged or non-stationary systems require adapted expressions.
- The values of the quantum extension are orders of magnitude; the Diósi–Penrose boundary depends on the mass distribution, and the proper-time differences quoted are upper bounds.

10 Conclusion

Expressing a system only through its time scales, then taking logarithms, turns dimensional analysis into linear algebra. Changes of unit become translations, power laws hyperplanes, regimes cells, eliminating a variable a kernel computation, and the deviation from a law a measurable distance whose logarithmic covariances propagate exactly. In this frame a reading rule can be proven: any relation that does not involve $\hbar$ can be written with the compactness X and the clock O alone, and the mass in Planck units Y becomes necessary as soon as $\hbar$ is involved among these same variables. The tool is not a theory; it is a common language and an instrument for classification, audit and design. A structure observed in this space could be called a candidate anomaly only if it is not an algebraic identity; if it does not come from variables reconstructed with the law being tested; if it exceeds the known errors and covariances; if it survives reasonable changes in the definition of the template; if it is recovered on data not used to build it; and if it appears in more than one class of systems or predicts a new observable. Such a regularity would be the subject of separate work.

Code and data availability

The associated Zenodo deposit contains: the Python module `empreintes.py` and its tests; `reproduire_papier.py`, which recomputes every number and figure; `verifier_equations.py`, which symbolically checks every equation (dimensions, factors, identities); the temporal atlas (`atlas_temporel.csv`); a self-contained interactive bench (HTML file); the L^AT_EX sources of the French and English versions. Everything is released under the Creative Commons BY-NC-SA 4.0 International licence.

Statement of assistance

The analysis, code and writing were carried out with the assistance of an artificial intelligence model (Claude, Anthropic). The author directed the work, checked the results and remains solely responsible for the content.

A Temporal atlas (excerpt)

Table 3: Temporal atlas (decimal logarithms). r_K : Kepler residual (9) for orbital clocks. Full atlas with absolute times and sources: file `atlas_temporel.csv`.

System	$\log X$	$\log Y$	$\log O$	r_K	Status of R
Electron (reduced Compton scale)	-44.76	22.38			defined
Proton (charge radius)	-38.83	19.11			measured
Hydrogen, Bohr orbit	-43.63	19.11	2.94		defined
Hydrogen, 21 cm line (maser)	-43.63	19.11	9.60		defined
Neutrons in superposition (COW 1975)	-52.45	19.11	4.97		measured
Rb atoms, 54 cm (2015)	-51.70	17.18	9.06		measured
Cs atoms held 20 s (2019)	-46.38	16.99	15.19		measured
25 kDa molecules (2019)	-42.94	14.72	12.94		order of magnitude
10^{-14} kg microdiamond (BMV proposal)	-37.53	6.34	12.48		proposed
Planck mass at the Planck length	0.00	0.00			defined
Earth surface (sidereal rotation)	-9.16	-32.44	6.61		measured
International Space Station	-9.19	-32.44	5.39	8.2×10^{-4}	measured
GPS satellite	-9.78	-32.44	5.69	9.8×10^{-5}	measured
Moon around the Earth	-10.93	-32.44	6.27	1.4×10^{-3}	measured
Sun surface (equatorial rotation)	-5.67	-37.96	5.98		measured
Mercury around the Sun	-7.59	-37.96	4.59	7.6×10^{-7}	measured (but $GM_\odot$ is calibrated on the orbits)
Earth around the Sun	-8.01	-37.96	4.80	-3.0×10^{-6}	measured (but $GM_\odot$ is calibrated on the orbits)
Jupiter around the Sun	-8.72	-37.96	5.16	-9.4×10^{-4}	measured (but $GM_\odot$ is calibrated on the orbits)
Neptune around the Sun	-9.48	-37.96	5.54	-6.3×10^{-4}	measured (but $GM_\odot$ is calibrated on the orbits)
Sirius B surface (white dwarf)	-3.57	-37.97			measured
Sirius A–B orbit	-8.81	-38.45	5.20	≈ 0	derived (Kepler)
PSR J0740+6620 surface (neutron star)	-0.61	-38.28	1.84		measured (NICER)
Crab pulsar surface	-0.76	-38.11	2.92		assumed (typical M, R)
PSR B1913+16 orbit	-5.67	-38.41	3.63	≈ 0	derived (Kepler)
Double pulsar J0737–3039 orbit	-5.36	-38.37	3.48	≈ 0	derived (Kepler)
Cygnus X-1 horizon	-0.30	-39.29			defined (Schwarzschild radius)
GW150914 at 40 Hz	-0.93	-39.78	0.96	≈ 0	derived (Kepler)
S2 around Sgr A*	-4.39	-44.59	2.99	≈ 0	derived (Kepler)
Sgr A* horizon	-0.30	-44.59			defined (Schwarzschild radius)
M87* horizon	-0.30	-47.77			defined (Schwarzschild radius)
Sun around the Galactic centre	-6.22	-48.98	3.91	≈ 0	circular: M derived from the same orbit
Hubble sphere today	-0.30	-60.63	-0.02		defined (c/H_0)
Hubble sphere at decoupling ($z \approx 1090$)	-0.30	-56.26	-0.22		defined ($c/H(z)$)

References

- [1] P. J. Mohr, D. B. Newell, B. N. Taylor, E. Tiesinga, “CODATA recommended values of the fundamental physical constants: 2022”, *Rev. Mod. Phys.* 97, 025002 (2025).
- [2] A. Prša et al., “Nominal values for selected solar and planetary quantities: IAU 2015 Resolution B3”, *Astron. J.* 152, 41 (2016).
- [3] M. Planck, “Über irreversible Strahlungsvorgänge”, *Sitzungsber. Preuss. Akad. Wiss.* 5, 440 (1899).
- [4] E. Buckingham, “On physically similar systems”, *Phys. Rev.* 4, 345 (1914).
- [5] J. L. Synge, *Relativity: The General Theory*, North-Holland (1960).
- [6] BIPM, *The International System of Units*, 9th edition (2019).
- [7] B. J. Carr, M. J. Rees, “The anthropic principle and the structure of the physical world”, *Nature* 278, 605 (1979).
- [8] C. H. Lineweaver, V. M. Patel, “All objects and some questions”, *Am. J. Phys.* 91 (10) (2023).
- [9] L. Diósi, “A universal master equation for the gravitational violation of quantum mechanics”, *Phys. Lett. A* 120, 377 (1987).
- [10] R. Penrose, “On gravity’s role in quantum state reduction”, *Gen. Rel. Grav.* 28, 581 (1996).
- [11] S. Bose et al., “Spin entanglement witness for quantum gravity”, *Phys. Rev. Lett.* 119, 240401 (2017).
- [12] C. Marletto, V. Vedral, “Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity”, *Phys. Rev. Lett.* 119, 240402 (2017).
- [13] M. Zych, F. Costa, I. Pikovski, Č. Brukner, “Quantum interferometric visibility as a witness of general relativistic proper time”, *Nat. Commun.* 2, 505 (2011).
- [14] A. Roura, “Gravitational redshift in quantum-clock interferometry”, *Phys. Rev. X* 10, 021014 (2020).
- [15] R. Colella, A. W. Overhauser, S. A. Werner, “Observation of gravitationally induced quantum interference”, *Phys. Rev. Lett.* 34, 1472 (1975).
- [16] T. Kovachy et al., “Quantum superposition at the half-metre scale”, *Nature* 528, 530 (2015).
- [17] V. Xu et al., “Probing gravity by holding atoms for 20 seconds”, *Science* 366, 745 (2019).
- [18] Y. Y. Fein et al., “Quantum superposition of molecules beyond 25 kDa”, *Nat. Phys.* 15, 1242 (2019).
- [19] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters”, *Astron. Astrophys.* 641, A6 (2020).
- [20] J. M. Weisberg, Y. Huang, “Relativistic measurements from timing the binary pulsar PSR B1913+16”, *Astrophys. J.* 829, 55 (2016).
- [21] M. Kramer et al., “Strong-field gravity tests with the double pulsar”, *Phys. Rev. X* 11, 041050 (2021).
- [22] GRAVITY Collaboration, “Detection of the Schwarzschild precession in the orbit of the star S2 near the Galactic centre massive black hole”, *Astron. Astrophys.* 636, L5 (2020).
- [23] LIGO Scientific and Virgo Collaborations, “Observation of gravitational waves from a binary black hole merger”, *Phys. Rev. Lett.* 116, 061102 (2016).
- [24] H. E. Bond et al., “The Sirius system and its astrophysical puzzles”, *Astrophys. J.* 840, 70 (2017).
- [25] E. Fonseca et al., “Refined mass and geometric measurements of the high-mass PSR J0740+6620”, *Astrophys. J. Lett.* 915, L12 (2021).
- [26] T. E. Riley et al., “A NICER view of the massive pulsar PSR J0740+6620”, *Astrophys. J. Lett.* 918, L27 (2021).

- [27] J. C. A. Miller-Jones et al., “Cygnus X-1 contains a 21-solar mass black hole”, *Science* 371, 1046 (2021).
- [28] Event Horizon Telescope Collaboration, “First M87 Event Horizon Telescope results. I”, *Astrophys. J. Lett.* 875, L1 (2019).